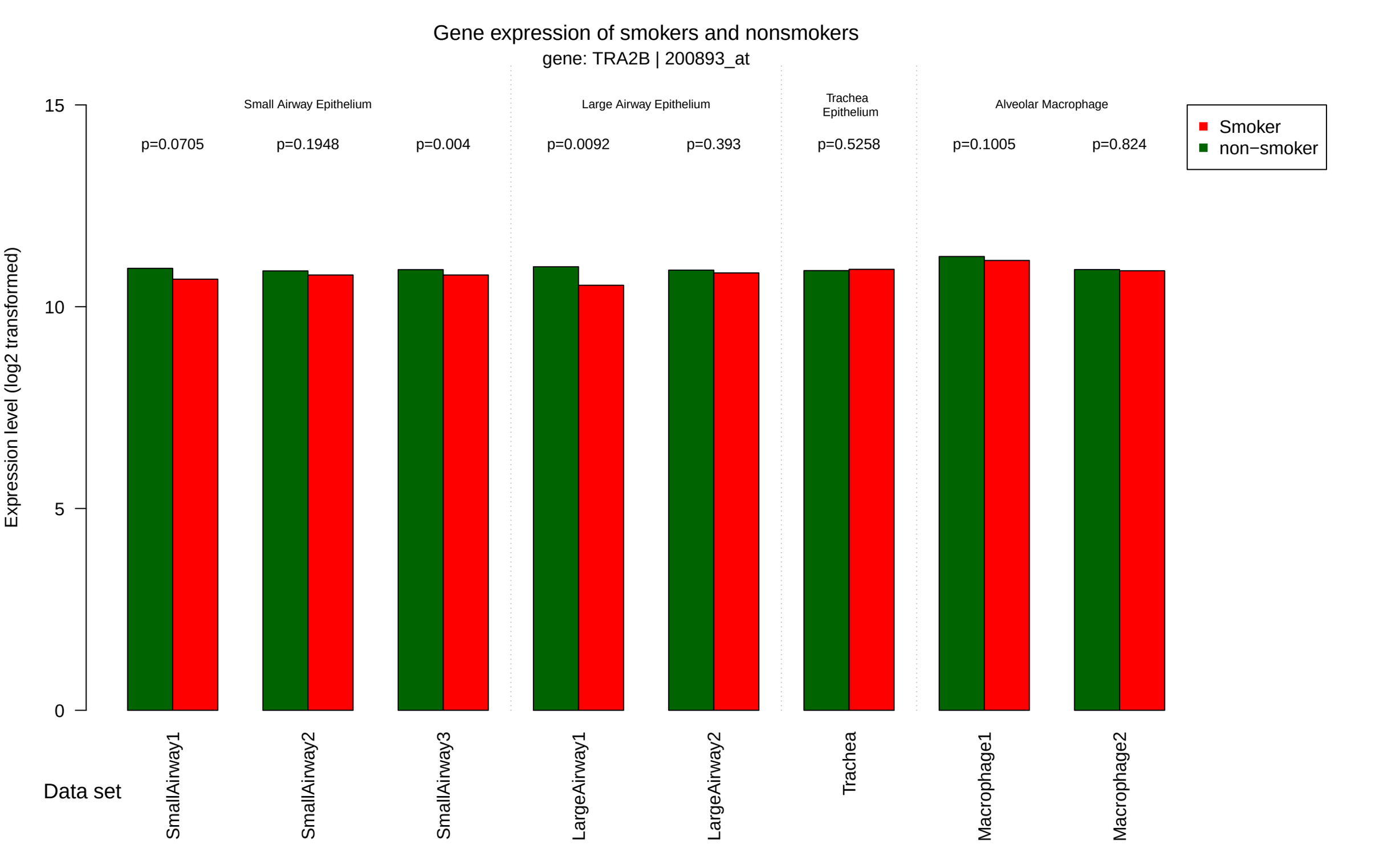
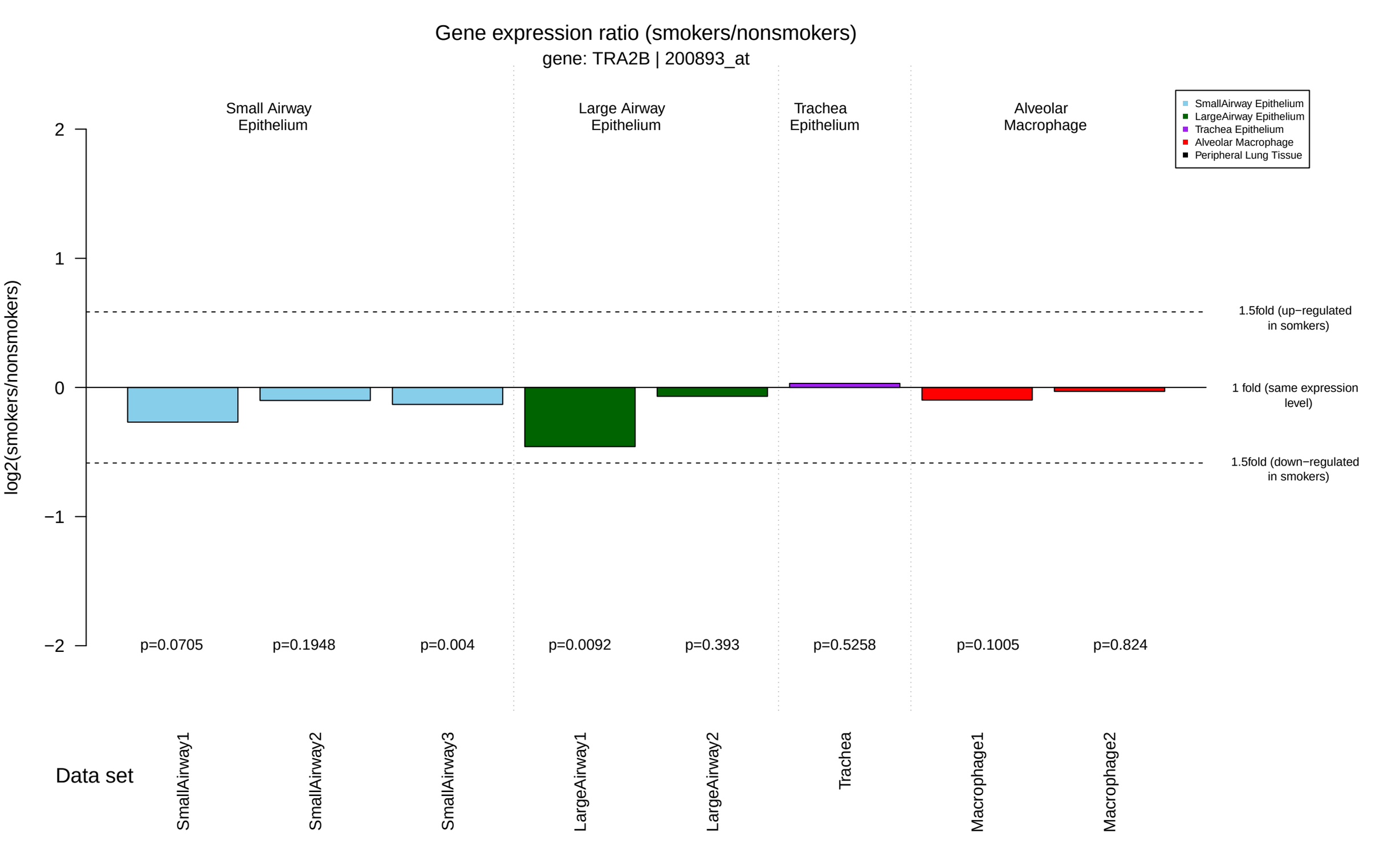
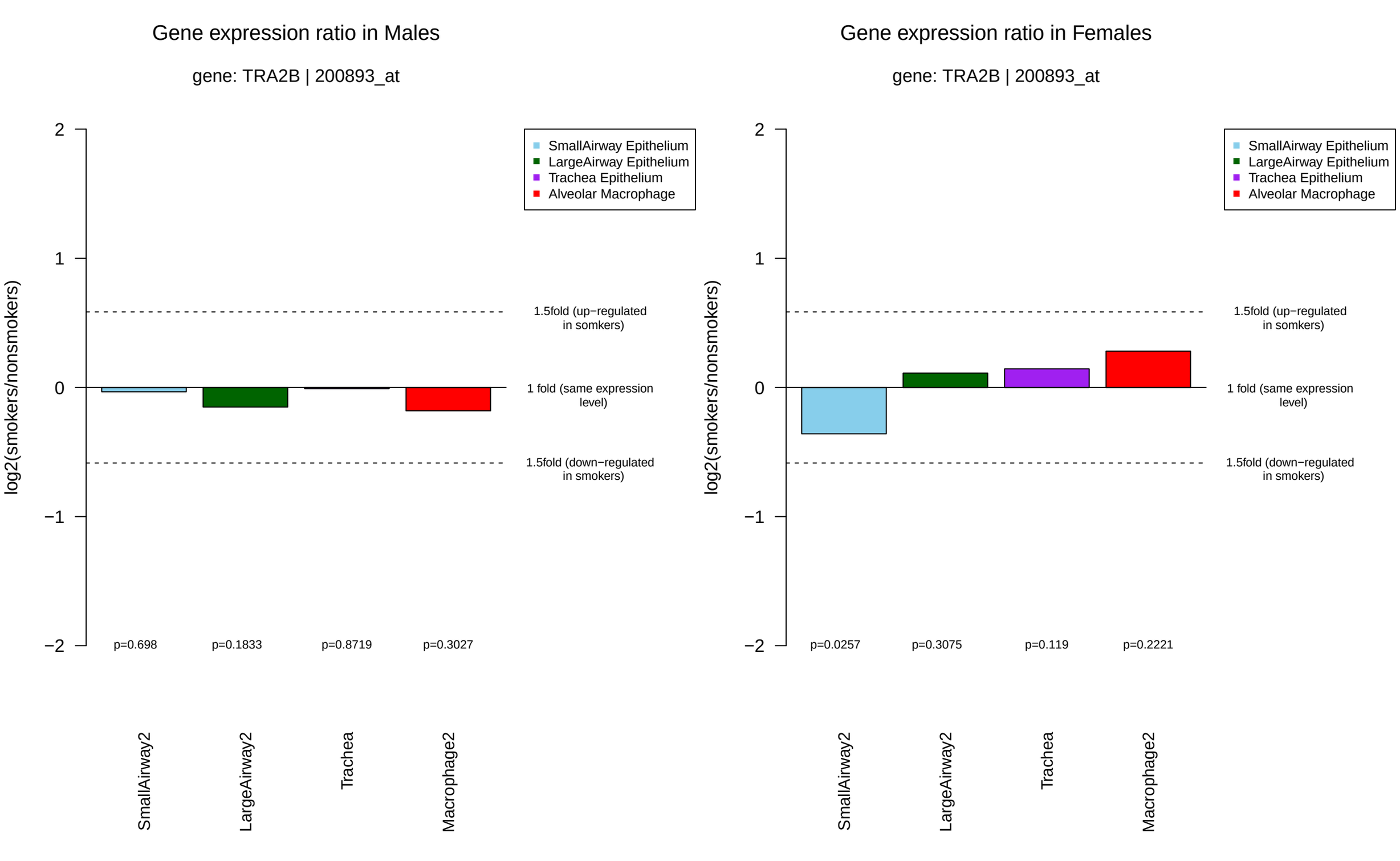
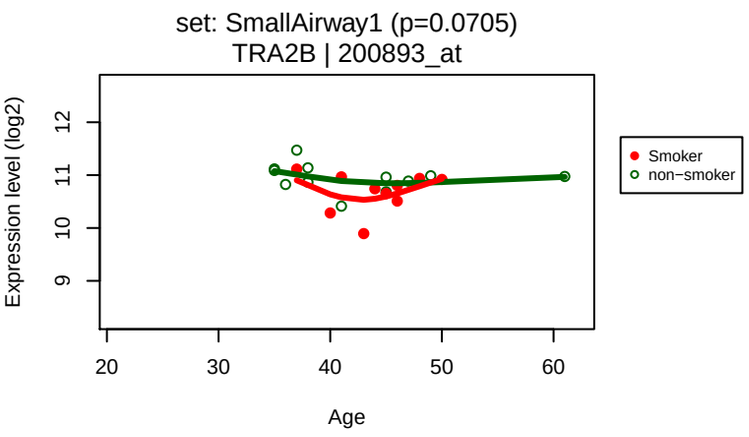
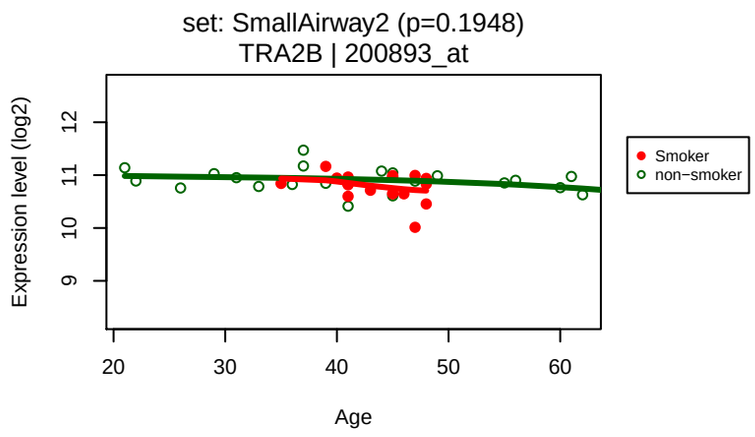
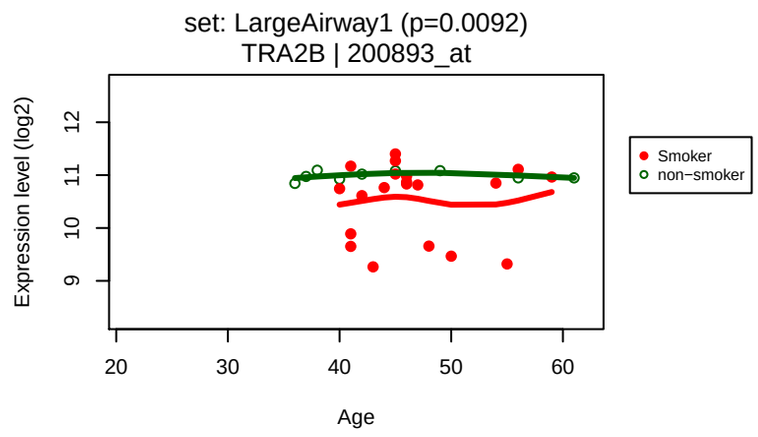

Gene expression of smokers and nonsmokers
gene: TRA2B | 200893_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0705
p=0.1948
p=0.004
p=0.0092
p=0.393
p=0.5258
p=0.1005
p=0.824
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

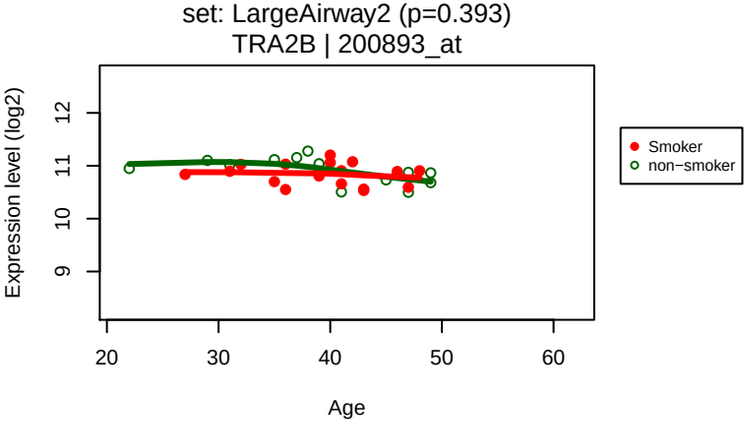

Gene expression ratio (smokers/nonsmokers)
gene: TRA2B | 200893_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0705
p=0.1948
p=0.004
p=0.0092
p=0.393
p=0.5258
p=0.1005
p=0.824
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

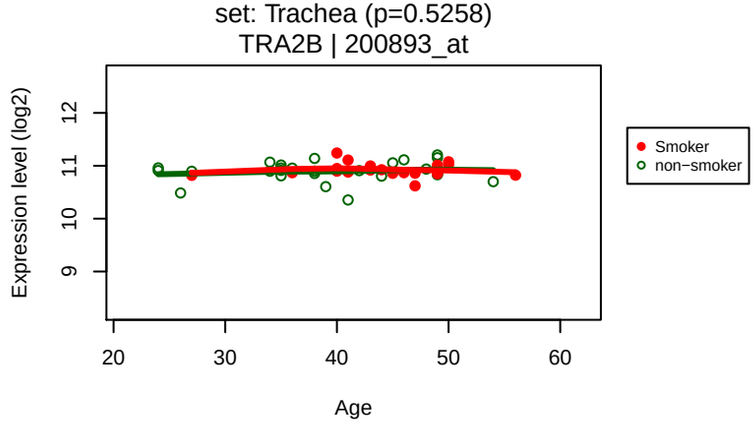

Gene expression ratio in Males
gene: TRA2B | 200893_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.698
p=0.1833
p=0.8719
p=0.3027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRA2B | 200893_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0257
p=0.3075
p=0.119
p=0.2221
SmallAirway2
LargeAirway2
Trachea
Macrophage2

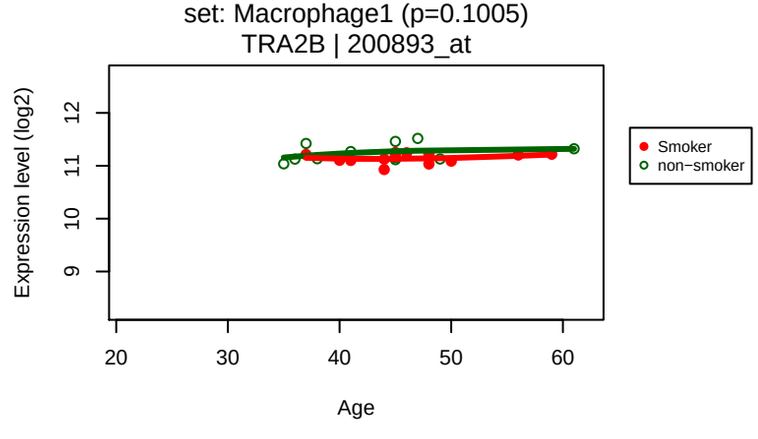

set: SmallAirway1 (p=0.0705)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker

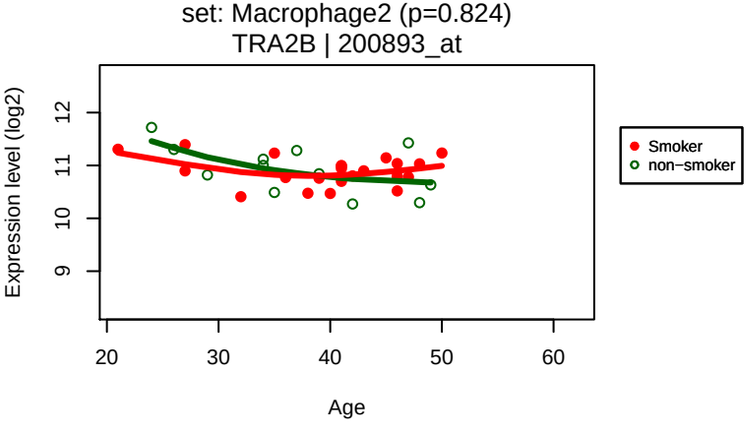

set: SmallAirway2 (p=0.1948)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker

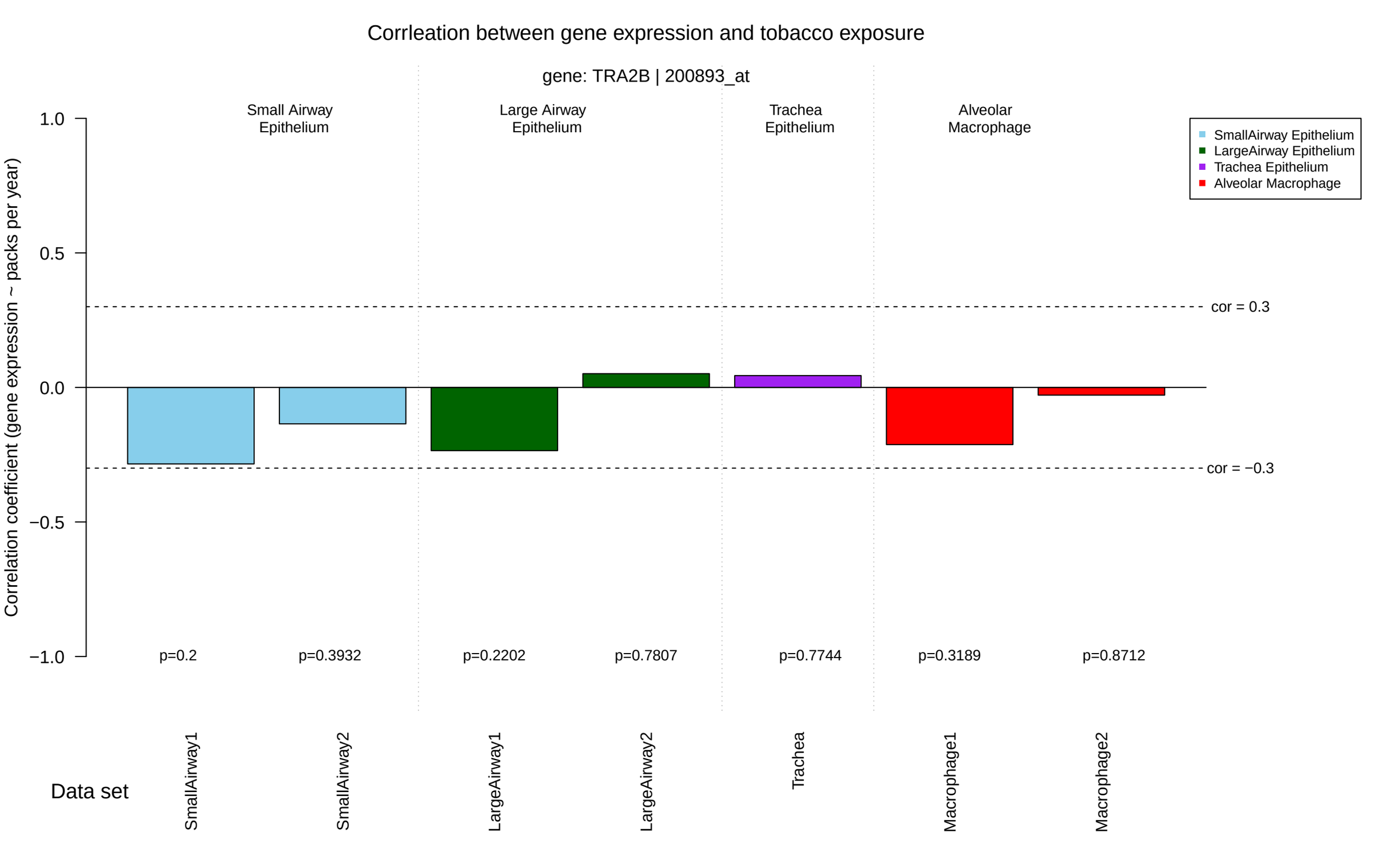

set: LargeAirway1 (p=0.0092)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.393)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.5258)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1005)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.824)
TRA2B | 200893_at
Expression level (log2)
9
10
11
12
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRA2B | 200893_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2
p=0.3932
p=0.2202
p=0.7807
p=0.7744
p=0.3189
p=0.8712
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2